Anurag Sharma

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Summary

AI/ML Engineer with ∼2 years building and **deploying** production LLM systems — **Agentic AI**, **RAG** and **Graph RAG** pipelines, and **LLM fine-tuning** (PEFT, **LoRA**) served on **AWS**. Currently builds insurance-operations automation at FBSPL: LLM-assisted policy checking, retrieval-first document extraction at **90–95% accuracy**, and a verification-gated UI automation engine with graph-based self-healing. **Deep learning** background spanning **CNN**/BiLSTM architectures through transformer fine-tuning and vLLM deployment.

Technical Skills

AI/ML & Deep Learning: Deep Learning, CNNs, BiLSTM, Self-Attention, Transformers, PyTorch, LLM Fine-Tuning (PEFT, LoRA, QLoRA), Model Deployment & Serving (vLLM), LLM Evaluation (RAGAS, DeepEval)

GenAI & Agentic AI: Agentic AI, Multi-Agent Systems, RAG, Graph RAG, Hybrid Search (BM25 + vector), Prompt Engineering, Tool/Function Calling, MCP, LangChain, LangGraph, Graphiti, Document Intelligence & OCR

Cloud & Deployment: AWS (EC2, S3, Bedrock), Docker, Kubernetes, CI/CD, Redis, Celery, Git

Languages & Data: Python, SQL, PostgreSQL, PostGIS, Neo4j, FalkorDB, MongoDB, Typesense, FAISS, Qdrant, Pinecone, ChromaDB

Backend & LLM APIs: FastAPI, REST APIs, Django, asyncpg, WebSockets, Rasa; Claude (Anthropic), OpenAI, AWS Bedrock, Gemini, Llama, Mistral, Qwen

Work Experience

AI/ML Engineer | *FBSPL* *Apr 2026 – Present*

Insurance operations automation. The agency management systems (Applied Epic, AMS360, EZLynx) have no API access, so document parsing and browser automation are the main paths.

- **Policy checking tool:** LLM-assisted comparison of issued policies against agency records. Reduced a 2–3 hour manual checklist to ∼45 minutes with ∼70% of items automated; handles chunked PDFs up to ∼300 pages. Surfaced a real policy-number mismatch between agency records and an issued excess policy.
- **Retrieval-first extraction:** native PDF text + **Typesense** hybrid search (BM25 + vector, Gemini embeddings) narrows context before LLM structuring — replacing a page-by-page multimodal approach and significantly lowering processing cost at similar accuracy on the tested workflow.
- **Document intelligence** (ACORD forms, dec pages, loss runs): layout-aware field linking that separates structural parsing from LLM-based field naming; hybrid **OCR** + LLM validation pipelines reach **90–95% extraction accuracy** on structured and semi-structured documents.
- **Verification-gated UI automation platform:** generalized end-to-end Applied Epic automation into an application-agnostic engine. Procedures are YAML runbooks synthesized from SOP documents and screen recordings, checked by a deterministic validator against a closed command vocabulary before they run.
- **Tiered resolution & graph self-healing:** each step resolves its target through ordered tiers (text anchor, cached selector, lexical match, LLM, vision), caching a selector only after its assertions pass and keying it on a screen fingerprint so procedures share what they learn; verified transitions form a state graph that routes a stray run back to the expected screen instead of failing.
- **Crew Support** (aviation crew-staffing): built a standalone geospatial **matchmaking service** (retrieve-then-rank) — hard eligibility filters (type rating, medical class, passport validity, PostGIS `ST_DWithin` radius, availability coverage) in SQL, with tunable weighted scoring in Python.
- **Explainable ranking:** weights in a `match_weight` table with a `ruleset_version` stamped per

run, **Bayesian shrinkage** on ratings, and an isolated read-only DB role so a scoring bug can only mis-order, never mis-assign crew.

- *Agnos:* **fine-tuned** Qwen2.5-Coder-1.5B-Instruct with **LoRA** as a HERMES tool-call router across 125 **agents**; fixed CJK garbage tokens (frozen `lm_head` → `modules_to_save`) to reach **99% routing / 100% priority, 0 garbage tokens** on 3K samples.
- *Agnos:* **deployed** on vLLM with the stock hermes parser (no sanitizers): ∼414 ms p50 end-to-end and ∼3,400 output tok/s sustained at 43.7 req/s / 100% SLA, benchmarked with GuideLLM.

AI/ML Engineer | *QSS Technosoft* *Feb 2026 – Apr 2026*

- Built a **Graph RAG** pipeline with **Graphiti** for temporal knowledge graphs (FalkorDB + Neo4j) enabling multi-hop retrieval across ad, podcast, and Spotify datasets; ran inference on **AWS Bedrock** (Claude, Titan) for NL-to-Cypher.
- Developed **scavenger-falkordb**, a custom **MCP** server exposing FalkorDB schema introspection and Cypher execution to Claude/LangGraph **agents**, with human-in-the-loop triple validation before graph commit.
- **MindCollect** (knowledge-graph analytics): hybrid **Graph RAG** layer combining a Neo4j knowledge graph with an HNSW vector index, an async ingestion pipeline (PDF/DOCX → chunking → embeddings → Neo4j + MongoDB) on S3, and a **multi-agent** framework (intent classification, context gathering, report generation).
- **Averisource** (legacy-code GenAI): feature for a mainframe-modernization platform that analyzes legacy **COBOL** and auto-generates documentation, business-rule summaries, and program-detail write-ups from source.

AI/ML Developer | *Plutos.One* *Sep 2025 – Feb 2026*

- Built an **LLM-based financial chatbot** presented at **Global Fintech Fest 2025**, handling structured financial queries and guided user flows.
- Contributed to a payment-network chatbot **deployed across 6 countries**, supporting multi-region users with real-time responses.
- Developed **FastAPI** backend services for model serving and request handling; improved response consistency and latency by restructuring prompts and the inference workflow.
- **WhatsApp visa chatbot:** multilingual Rasa assistant integrated with Gupshup, using Redis/Celery for async handling and **Docker** for deployment; load-tested to 5K+ users.

Software Developer Intern | *DigiMonk Technologies* *Jun 2024 – Nov 2024*

- Built an AI Sales Dialer Agent (Vapi.ai + Twilio) boosting lead conversion 25%; REST APIs cut manual work 40%.

Education & Research

M.Tech in Computer Science *(pursuing)* — Shriram College of Engineering & Management

Thesis — real-time gesture recognition: MediaPipe Holistic → parallel **1D-CNN** encoders → **BiLSTM** with self-attention, evaluated on SHREC'17 and a self-recorded gesture dataset. Review paper submitted to IJSREM (under review).

B.Tech in Information Technology — ITM University, Gwalior 2020–2024 CGPA: 8.0

Full Stack Data Science — Naresh IT, Hyderabad Jan 2025 – Jun 2025

Certifications

Cambridge Business English Certificate 2024